

ClinicConnect HMS - Complete Development Roadmap

This document summarizes the complete Software Development Life Cycle (SDLC) roadmap for the ClinicConnect Hospital Management System project.

Phase 0 – Project Planning

Problem statement, feature discussion, user roles, SRS, architecture, pharmacy integration, notifications, chat, appointment flow, landing page discussion.

Phase 1 – Project Setup

GitHub repository, branching strategy, React + Vite frontend, Node.js + Express backend, MongoDB Atlas, environment variables, folder structure, README.

Phase 2 – UI/UX

Landing website (Navbar, Hero, About, Departments, Doctors, Facilities, Testimonials, Contact, Footer), public pages, dashboards for Patient, Doctor, Receptionist, Pharmacist and Admin.

Phase 3 – Authentication

JWT authentication, password hashing, role-based authorization, login, registration, forgot password, OTP.

Phase 4 – Master Data

Hospital, departments, doctors, receptionists, pharmacists and patients. Admin CRUD.

Phase 5 – Appointment Module

Appointment booking flow, doctor schedules, reception management, reschedule and cancellation.

Phase 6 – Consultation

Symptoms, diagnosis, notes and prescription management.

Phase 7 – Pharmacy

Prescription processing, billing, invoices and inventory updates.

Phase 8 – Patient Portal

Appointments, reports, prescriptions, invoices, profile, notifications and chat.

Phase 9 – Chat Module

Doctor-patient real-time chat using Socket.IO with attachments and read status.

Phase 10 – Notifications

Appointment reminders, doctor availability, medicine reminders, billing and prescription notifications.

Phase 11 – Analytics

Admin and doctor dashboards with appointments, revenue, patients and inventory statistics.

Phase 12 – AI Features

AI symptom checker, appointment assistant, prescription OCR, chatbot and predictive analytics.

Phase 13 – Testing

Backend API testing, frontend validation, responsive testing and integration testing.

Phase 14 – Deployment

Frontend on Vercel, backend on Render, MongoDB Atlas, optional custom domain.

Phase 15 – Documentation

README, architecture diagram, ER diagram, API documentation, SRS, project report, presentation and demo script.

Suggested Sprint Plan

Sprint 1: Setup + Landing Page

Sprint 2: Authentication

Sprint 3: Master Data

Sprint 4: Appointment Module

Sprint 5: Consultation + Prescription

Sprint 6: Pharmacy + Billing

Sprint 7: Patient Portal

Sprint 8: Chat + Notifications

Sprint 9: Analytics

Sprint 10: Testing + Deployment

Sprint 11: AI Features + Documentation

Git Workflow

main ← develop ← feature branches (auth, landing-page, appointment, pharmacy, chat, dashboard). Never push directly to main.

Feature Development Workflow

Requirement → Database Design → API Design → Backend → API Testing → Frontend → Integration → Testing → GitHub Push → Pull Request → Merge into develop.

Team Distribution

Project Lead: Backend architecture, authentication, database, API integration, GitHub management.

Member 1: Landing page, patient UI, dashboard UI, responsiveness.

Member 2: Doctor module, pharmacy module, integration and testing.